Gas Dynamics Before Scientific Machine Learning

Exact Relations, Branch-Aware Inversion, and Evidence-Based Surrogates

Week 8 Extension Lecture Notes

M&I-ENG 690A – AI in Fluid Mechanics

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Central question. When an exact compressible-flow relation or a bracketed root is already available, what evidence is required before a neural surrogate is scientifically useful?

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1 Position in the Course

Weeks 1–7 establish numerical validation, kinetic sampling, supervised learning, operator learning, controlled research protocols, and unsteady cylinder wakes. Week 8 adds compressible-flow physics only after students have learned to separate a numerical label, a validation target, and a learned approximation.

This module connects three public repositories while preserving their distinct scientific roles:

Source	Role in Week 8	Claim boundary
Compressible-flow book code	nine chapter-aligned classical notebooks	teaching solvers; analytical and conservation-law checks remain primary
GasDynBench	five exact/inverse benchmarks, matched baselines, and frozen evidence	interpretable inverse benchmark; not a replacement for CFD
SU2 diamond-airfoil code	bridge to multidimensional shock-capturing CFD	one qualified $\alpha = 0$ Euler teaching case; eight other cases remain unverified

The teaching narrative is aligned with the author-supplied revised manuscript *Physics-Guided Neural Surrogates for Canonical Compressible Thermal-Fluid Relations* (manuscript `AITF-D-26-00044R1`). A revision identifier is not evidence of acceptance; executable claims remain tied to the frozen public repository, machine-readable tables, and tests.

Learning checkpoint

By the end of Week 8, a student should be able to:

1. identify the governing relation, physical domain, and branches before defining an ML input/output map;
2. compute Rayleigh, Fanno, oblique-shock, nozzle-shock, and shock-tube references;
3. distinguish algebraic evaluation, bracketed inversion, ODE integration, and multidimensional CFD;
4. demonstrate why hiding branch identity produces an ill-posed regression;
5. compare a physics-guided MLP with exact evaluation and interpolation under matched information;
6. test physical bounds, closure residuals, edge generalization, storage, and timing; and
7. state precisely when the neural model helps and when a classical method remains preferable.

2 Compressible-Flow Foundations

For a calorically perfect gas,

$$p = \rho R T, \qquad a = \sqrt{\gamma R T}, \qquad \mathrm{Ma} = \frac{|\boldsymbol{u}|}{a}. \tag{1}$$

The inviscid conservation laws in differential form are

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{2}$$

$$\frac{\partial (\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + p\boldsymbol{I}) = 0, \tag{3}$$

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot [(\rho E + p)\boldsymbol{u}] = 0. \tag{4}$$

For one-dimensional isentropic changes, stagnation properties obey

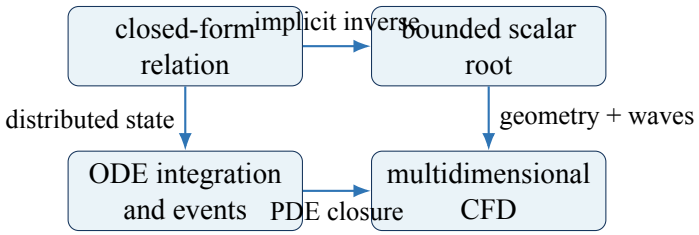
$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}\mathrm{Ma}^2, \tag{5}$$

$$\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2}\mathrm{Ma}^2\right)^{\gamma/(\gamma-1)}. \tag{6}$$

Important distinction

A relation can be numerically smooth and still be non-invertible without a branch label. Sonic points, shock detachment, internal-shock bounds, and contact compatibility define the topology of the learning problem.

2.1 Four computational levels



The baseline must match the level. A neural network should not be compared only with a slow generic optimizer when a closed-form relation or accurate lookup table exists.

3 Five Canonical Reference Families

Figure 1 shows the exact maps used before learning.

3.1 Rayleigh flow: heat transfer and sonic choking

For steady, frictionless, constant-area flow with heat transfer, the sonic reference ratios include

$$\frac{T}{T^*} = \frac{(1+\gamma)^2\mathrm{Ma}^2}{(1+\gamma\mathrm{Ma}^2)^2}, \qquad \frac{p}{p^*} = \frac{1+\gamma}{1+\gamma\mathrm{Ma}^2}, \tag{7}$$

$$\frac{T_0}{T_0^*} = \frac{T}{T^*}\frac{1 + \frac{1}{2}(\gamma-1)\mathrm{Ma}^2}{\frac{1}{2}(\gamma+1)}. \tag{8}$$

The inverse $T_0/T_0^* \mapsto \mathrm{Ma}$ has subsonic and supersonic roots. The static ratio T/T^* reaches its maximum at $\mathrm{Ma} = 1/\sqrt{\gamma}$, so splitting that inverse at $\mathrm{Ma} = 1$ is inconsistent. Week 8 therefore uses the stagnation temperature ratio for a clean sonic branch split.

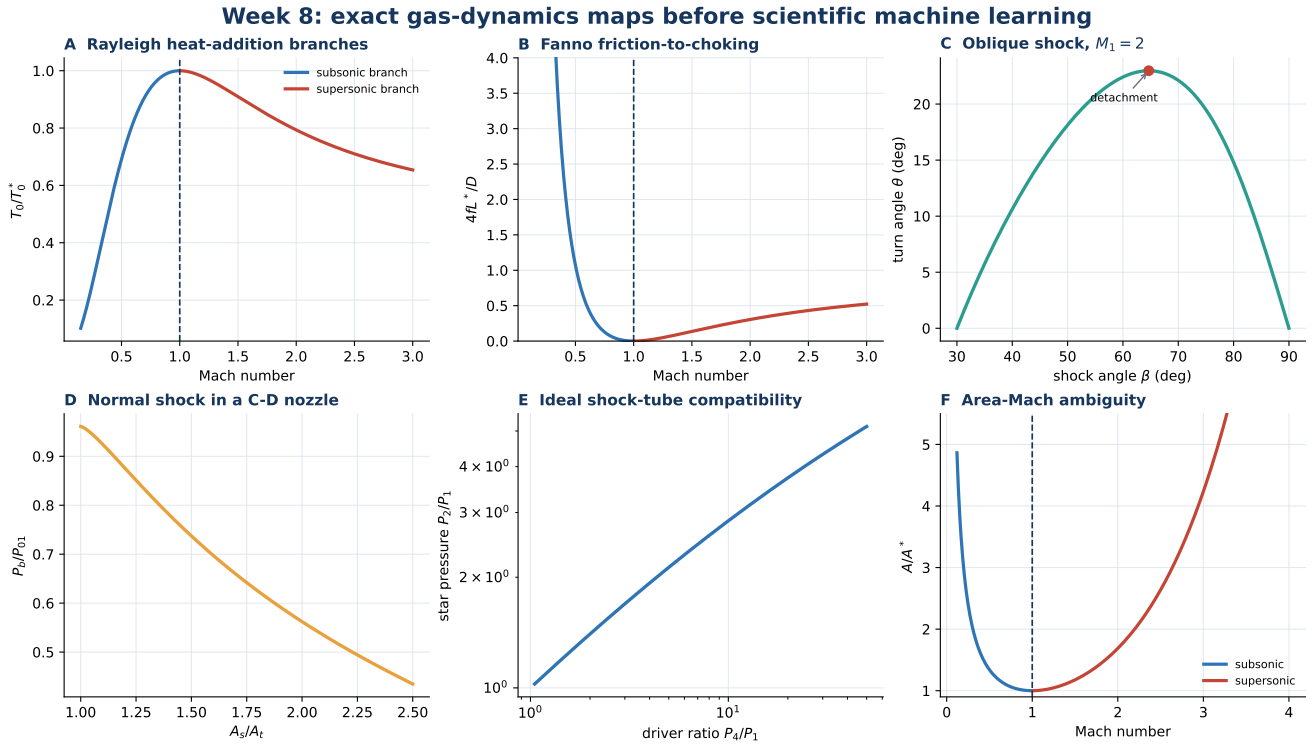


Figure 1: Exact gas-dynamics maps generated by the FlowMLLab Week-8 module. Dashed sonic lines and turning-point geometry expose the branches that an inverse model must retain.

3.2 Fanno flow: friction length to choking

For adiabatic constant-area flow with friction,

$$\frac{4fL^*}{D} = \frac{1-\mathrm{Ma}^2}{\gamma \mathrm{Ma}^2} + \frac{\gamma+1}{2\gamma} \ln\left[\frac{(\gamma+1)\mathrm{Ma}^2}{2+(\gamma-1)\mathrm{Ma}^2}\right]. \tag{9}$$

The quantity is nonnegative and vanishes at the sonic state. The subsonic branch has an unbounded upstream friction length as $\mathrm{Ma} \to 0$, while the supersonic branch approaches a finite limit. A bounded output transform can enforce nonnegativity, but it cannot replace a domain check.

3.3 Oblique shocks: weak and strong attached roots

The theta-beta-M relation is

$$\tan\theta = 2\cot\beta \, \frac{\mathrm{Ma}_1^2 \sin^2\beta - 1}{\mathrm{Ma}_1^2(\gamma + \cos 2\beta) + 2}. \tag{10}$$

For $0 < \theta < \theta_{\max}$, weak and strong roots exist on opposite sides of the detachment maximum. At $\theta_{\max}$ they coalesce. Above it, no attached root exists. A single naive regressor trained on both roots learns their average, which is generally neither physical branch.

3.4 Normal shock inside a converging-diverging nozzle

The isentropic area relation is

$$\frac{A}{A^*} = \frac{1}{\mathrm{Ma}} \left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}\mathrm{Ma}^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}. \tag{11}$$

Worked inverse: branch before regressor. For $\gamma = 1.4$ and $A/A^* = 1.6875$, define the dimensionless residual

$$F(M) = \frac{1}{M}\left[\frac{5}{6}(1 + 0.2M^2)\right]^3 - 1.6875.$$

On the supersonic bracket $[1, 3]$, $F(1) = -0.6875$ and $F(3) \simeq 2.54707$. Its first bisection midpoint is $M = 2$, and

$$\frac{A}{A^*} = \frac{1}{2}\left[\frac{5}{6}(1 + 0.2 \times 4)\right]^3 = \frac{1}{2}(1.5)^3 = 1.6875.$$

Thus the forward residual is zero. This convenient exact midpoint is specific to the chosen example; an arbitrary target requires further iterations. On the subsonic bracket $[0.1, 0.9]$, the residual changes from 4.13433 to -0.678637, giving the second root $M \simeq 0.3722444862$. Both roots have the same area ratio. The flow region and shock or boundary conditions select the branch; averaging the roots does not solve the inverse. This is one isentropic segment, not the complete internal-shock calculation.

Self-check (10 points): bracket signs and midpoint substitution (4), both roots and physical branch information (4), residual plus branch check (2). For $A/A^* = 1$ the branches meet at $M = 1$: an endpoint root need not have a sign-changing bracket on an open branch. All quantities here are dimensionless.

It also has subsonic and supersonic branches. An internal normal shock maps A_s/A_t to p_b/p_{01} through upstream isentropic acceleration, the normal shock jump, and downstream subsonic diffusion. The inverse is bracketed inside

$$1 < \frac{A_s}{A_t} < \frac{A_e}{A_t}. \tag{12}$$

The geometry supplies a hard output bound for both root solvers and learned models.

3.5 Shock-tube pressure compatibility

The incident shock and driver expansion must produce the same pressure and velocity at the contact surface. The equal-gas relation is implicit in p_2/p_1 and is solved with a bracketed root. The generalized benchmark varies

$$\left(\frac{p_4}{p_1}, \frac{T_4}{T_1}, \gamma_1, \gamma_4, \frac{R_4}{R_1}\right) \longmapsto \frac{p_2}{p_1}. \tag{13}$$

The residual of the compatibility equation remains a required diagnostic after any neural approximation.

Learning checkpoint

For each family, write down: input, output, branch or bound, exact/numerical reference, and a forward-substitution residual. If one item is missing, the ML problem is not ready to train.

4 Beyond the Five Benchmarks

The nine classical notebooks extend the physics without pretending that every problem needs a neural surrogate:

- shock-polar diagrams organize admissible post-shock velocity states;
- colliding oblique shocks require pressure compatibility and a slip line;
- interacting unsteady shocks require a corrected nonlinear pressure-ratio relation;
- Taylor–Maccoll conical flow requires ODE integration from the shock and an event that locates the cone surface; and
- parameter sweeps must monitor event failure and admissibility, not only integrator status.

These notebooks are linked from Week-8 Lab 1 and remain authoritative in the compressible-flow book-code repository.

5 From Exact Physics to a Learned Inverse

5.1 Dataset construction

An auditable inverse benchmark follows this sequence:

1. declare the physical domain and every branch;
2. sample complete physical states from the exact forward relation;
3. compute inverse targets with a bracketed solver and residual tolerance;
4. split complete states into development, validation, and blind sets;
5. fit scalers on development data only;
6. train branch-specific or explicitly branch-conditioned models;
7. freeze architecture, transforms, and gates before opening blind data; and
8. evaluate global error, pointwise error, physical bounds, and closure residuals.

5.2 Hard transforms and interpretable constraints

Useful transforms encode known structure. Examples include a logistic map for the nozzle location,

$$\frac{A_s}{A_t} = 1 + \left(\frac{A_e}{A_t} - 1\right)\sigma(z), \tag{14}$$

and a nonnegative Fanno construction,

$$\frac{4fL^*}{D} = (\mathrm{Ma} - 1)^2 \exp[g(\mathrm{Ma})]. \tag{15}$$

These transforms enforce a bound or limit exactly. They do not prove accuracy, generalization, entropy consistency, or superiority to interpolation.

5.3 Five ways the article inserts physics

The manuscript uses "physics-guided" in a broader sense than a strict PINN: the loss does not enforce the full compressible-flow PDE residual. Instead, physical knowledge changes the supervised learning problem itself.

Design device	Example	What it prevents
Branch decomposition	Rayleigh/Fanno inverse experts	averaging incompatible Mach roots
Sonic coordinate + positive transform	$(\mathrm{Ma}-1)^2\exp[g(\mathrm{Ma})]$ for Fanno	negative friction length and broken sonic limit
Physics-informed features	Ma, $\log \mathrm{Ma}$, Ma^2 for Rayleigh	poor conditioning over wide Mach ranges
Exact boundary envelope	oblique weak/strong endpoints	detached or misordered attached-shock outputs
Hybrid analytical reconstruction	shock-tube implicit pressure step	learning discontinuous fields that exact wave relations reconstruct

The common reported training contract uses Tanh activations, L-BFGS, an L_2 weight penalty of 10^{-8}, tolerance 10^{-10}, at most 700 iterations and 50,000 function evaluations. Hyperparameters are frozen before opening the blind grids; seed 11 supplies the main tables, while seeds 11, 29, and 47 measure optimizer/training variability rather than calibrated uncertainty.

5.4 Why two experts are not optional

Suppose identical scalar inputs x correspond to y_s and y_f on slow and fast branches. Mean-squared training without a branch label drives a deterministic regressor toward

$$\widehat{y}(x) \approx \mathbb{E}[y \mid x], \tag{16}$$

which lies between the roots. Week-8 Lab 2 constructs this failure for the Rayleigh inverse, then replaces the ill-posed map with two declared experts.

6 What the Retained Evidence Shows

The primary blind-test results are copied byte-for-byte from `GasDynamicsSciML` commit `374431a`; the complete source SHA is recorded in `results/gas_dynamics_week8/provenance.json`.

Inverse problem	Relative L_2	Relative L_∞	Physical coverage
Rayleigh	0.1642%	0.2521%	100%
Fanno	0.3431%	0.3761%	100%
Oblique shock	0.0799%	0.4543%	100%
Nozzle shock location	0.1936%	0.4756%	100%
Shock-tube implicit step	0.0949%	0.4209%	100%

All five pass the declared primary relative-L_2 gate of 0.35%. This is a statement about the retained domains and split, not unrestricted extrapolation.

6.1 The matched interpolation baseline

For Rayleigh, Fanno, and the shock-tube table, interpolation is substantially more accurate wherever it returns a value. It is also faster for covered one-dimensional tables. Oblique and nozzle inverses favor the physics-guided MLP in the retained comparison, and the MLP has complete coverage for all five tasks.

Decision rule

Use the exact formula when it is cheap. Use interpolation for a covered low-dimensional table when accuracy, coverage, storage, and derivatives satisfy the application. Consider a bounded neural surrogate when repeated implicit solves, dimensionality, compact storage, or differentiable evaluation create a measured advantage.

6.2 Edge generalization is a separate experiment

Training on interior parameter boxes and testing only omitted low/high bands raises relative-L_2 error to 3.41% for Rayleigh, 4.30% for Fanno, 0.755% for oblique shocks, 1.06% for the nozzle, and 0.432% for the shock tube. These are in-domain edge tests. Calling them unrestricted extrapolation would still be incorrect.

6.3 Dimensional scaling under matched data budgets

With about 4096 states, a regular grid has 64 points per axis in two dimensions but only 5 per axis in five dimensions. At five inputs, interpolation has 4.88496% relative-L_2 error, while the bounded MLP has 0.177106%. Online interpolation remains slightly faster; the demonstrated advantage is compact accuracy under sparse high-dimensional sampling.

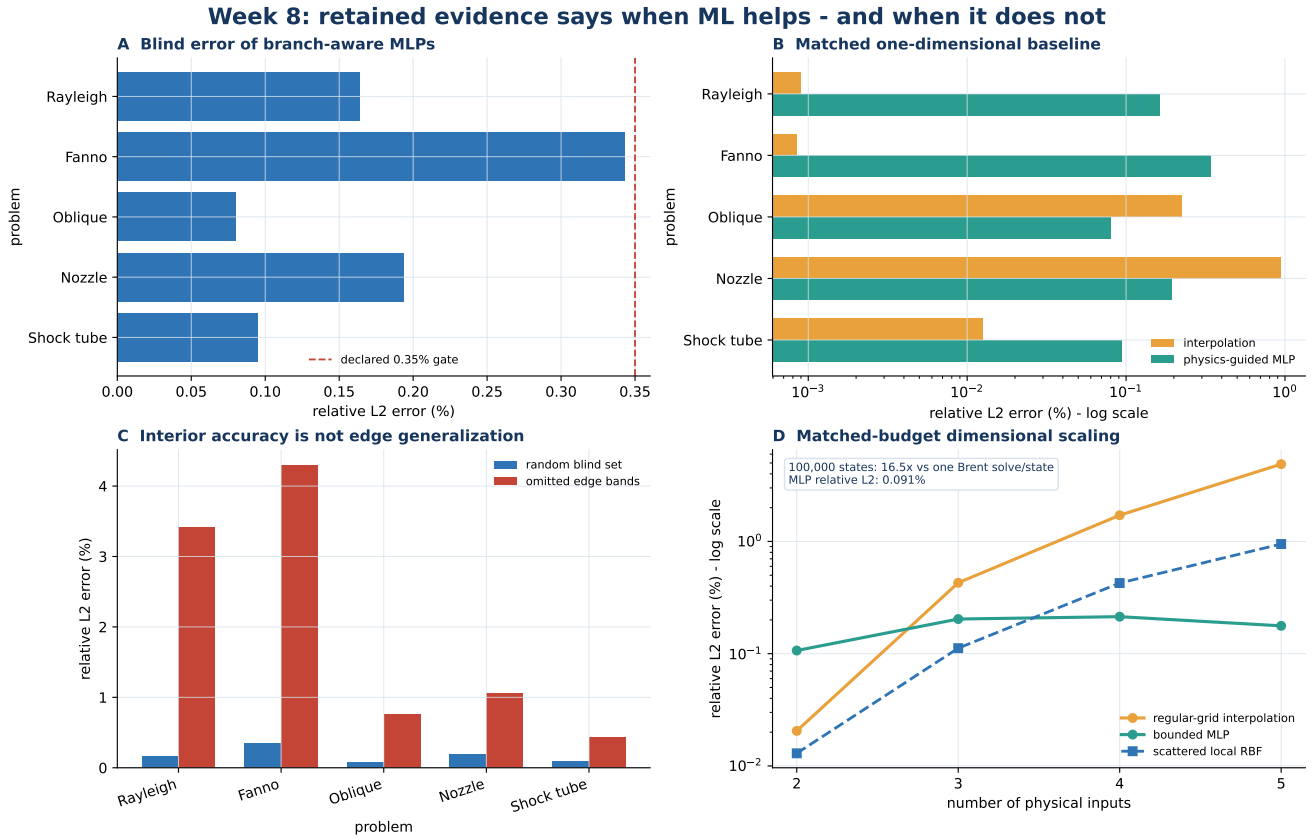


Figure 2: Retained evidence. Panel B shows why "ML wins" is not a general conclusion on one-dimensional tables. Panel C separates ordinary blind accuracy from omitted-edge testing. Panel D shows the matched-budget dimensional crossover.

That regular-grid comparison alone is not a sufficient baseline because the MLP receives 4096 Latin-hypercube states while the grid receives only five locations per coordinate. FlowMLLab therefore adds a local thin-plate RBF using the same 4096 scattered training states and the same 2048 blind states. Its errors for two through five inputs are 0.0130%, 0.1120%, 0.4260%, and 0.9487%. Thus the fairer scattered baseline closes most of the apparent gap; the five-input MLP remains more accurate, but by a factor of about 5.4 rather than the factor of 27.6 suggested by the regular grid alone. This added audit is FlowMLLab teaching evidence, not a value transcribed from the manuscript.

6.4 Application-scale timing and derivatives

For 100,000 generalized shock-tube states in one single-thread process, the MLP is 16.53 times faster than one Brent solve per state with 0.09148% relative-L_2 error. In a nine-case nozzle inverse-design audit, safeguarded Newton iteration converges in at most six steps, and the analytical network Jacobian agrees with centered finite differences to 1.37×10^{-9} relative difference.

Important distinction

Timing evidence is valid only for the recorded workload, implementation, hardware environment, thread limits, warm-up, and statistic. It does not imply that an MLP is faster than a closed-form formula or a prepared interpolation table.

7 Physical Diagnostics Beyond Error Norms

Global errors must be accompanied by problem-specific checks:

Problem	Required diagnostic	Failure meaning
Rayleigh	entropy trend and sonic peak location	wrong thermodynamic direction or shifted choke state
Fanno	nonnegative $4fL^*/D$ and sonic limit	invalid friction length or broken limiting behavior
Oblique shock	weak/strong branch validity and endpoints	averaged or detached state represented as attached
Nozzle	$1 < A_s/A_t < A_e/A_t$ and backpressure closure	shock outside nozzle or inverse mismatch
Shock tube	$p_2/p_1 > 1$ and compatibility residual	star state violates wave matching

The retained evidence stores these diagnostics in CSV rather than inferring them from visually smooth plots.

Hard bounds certify only the encoded property. For example, a valid nozzle location does not prove backpressure closure, and a positive shock-tube pressure does not prove contact compatibility. The manuscript therefore reports both validity rates and residual discrepancies; students must keep those columns separate.

8 Bridge to Multidimensional CFD

The Mach-3 diamond-airfoil repository contains Euler, laminar Navier–Stokes, and SST RANS configurations at $\alpha = 0^\circ, 4^\circ, 8^\circ$. Its two-stage workflow uses robust first-order initialization followed by second-order MUSCL mean-flow reconstruction. The wrapper checks more than the SU2 exit code:

1. fresh output and complete provenance;
2. residual history and declared reduction;
3. at least 200 force samples in the final window;
4. force-window stability and expected ranges;
5. nonphysical-point and reconstructed-state warnings;
6. symmetry and shock-angle metrics from native-grid connectivity; and
7. wall-resolution metrics where applicable.

At the frozen Week-8 commit, the sharp-wall `euler_alpha0` case is a *qualified teaching reference*: its forces, symmetry, shock angle, and physicality checks are enforced, while a plateaued density residual remains an explicit warning. The other eight cases have unresolved acceptance entries and remain unverified.

Evidence gate

No unverified SU2 field enters the Week-8 training data. A future student project may qualify one additional case only by predeclaring its acceptance ranges, running a mesh/time or iteration study as appropriate, and preserving failed variants.

9 Two-Lab Teaching Plan

9.1 Lab 1: exact gas dynamics before ML

Students execute five references, recover both roots where applicable, close the inverse by forward substitution, and classify all nine chapter notebooks by numerical method. The exit ticket is a reference-solver evidence card.

9.2 Lab 2: when should a surrogate be neural?

Students first expose the hidden-branch failure with a small deterministic MLP. They then inspect the five retained blind results, interpolation coverage, omitted-edge errors, dimensional scaling, the 100,000-state workload, and physical diagnostics. The exit ticket requires an exact/interpolation/MLP decision.

Assessment item	Required evidence
Problem definition	nondimensional input/output, domain, branch, and bound
Reference	equation or solver, tolerance, forward-substitution residual
Baseline	matched interpolation or root solve with equal information
Learning protocol	complete states, development-only selection, frozen gate
Physics	admissibility, entropy/closure/limit diagnostic
Generalization	blind split and a separately named edge test
Decision	limited claim plus one condition under which the model is rejected

10 Reproducibility Contract

The FlowMLLab copy records source commits, continuous-integration runs, and SHA-256 hashes for each retained CSV/JSON file. The compact Week-8 check is

```
python qa/build_week8_gas_dynamics_figures.py
python -m unittest tests.test_gas_dynamics
python qa/validate_course_release.py
```

The full research regeneration remains in `GasDynamicsSciML`:

```
git checkout 374431a1033138f56e2752bf8bbf9b75a454d80c
python -m pip install -e .
python -m unittest discover -s tests -v
gasdynbench --output results/revision
```

Record Python version, thread limits, seeds, workload, and evidence hashes before comparing a regenerated result with the retained tables.

11 Conclusions

1. Exact physics defines the dataset, branch structure, bounds, and residuals.
2. A branch-hidden inverse is mathematically ill-posed; more training cannot repair it.
3. Interpolation is a required baseline and often wins on covered 1-D tables.
4. Neural value must be demonstrated through coverage, dimensional scaling, repeated implicit workloads, compactness, or derivatives – not assumed.

5. Edge tests, physical diagnostics, and timing protocols are separate from ordinary blind error.
6. Multidimensional CFD requires its own verification and validation chain before becoming an ML label.

Repository and Reading Guide

- E. Roohi, *Introduction to Compressible Flows*, computational companion repository.
- E. Roohi, *GasDynamicsSciML / GasDynBench*, scientific-ML repository.
- E. Roohi, *Physics-Guided Neural Surrogates for Canonical Compressible Thermal-Fluid Relations*, revised manuscript `AITF-D-26-00044R1`, 2026.
- E. Roohi, *SU2 Diamond-Airfoil Verification*, teaching-CFD repository.
- J. D. Anderson, *Modern Compressible Flow*, McGraw-Hill.
- A. H. Shapiro, *The Dynamics and Thermodynamics of Compressible Fluid Flow*, Ronald Press.
- P. A. Thompson, *Compressible-Fluid Dynamics*, McGraw-Hill.